

Set 11: Periodic trends

Notes

The Periodic Table was developed over many years and ordered all known elements into an arrangement that made it possible to make predictions about properties both physical and chemical.

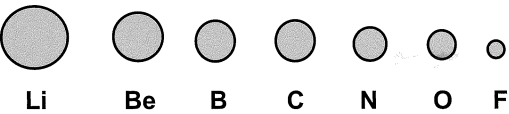
Going from left to right the horizontal rows are called periods. The vertical columns are called groups.

Periodic trends and patterns:

When going from left to right across a period in the Periodic Table:

- Atomic radii decrease
- Electronegativity increases
- Number of valence electrons increases
- Ionisation energy increases
- Electrical and thermal conductivity decrease
- Elements change from metal to metalloid to non-metal
- Bonding in elements changes from metallic to covalent network to covalent molecular

Set 11: Exercises

1. What general trends or patterns are there when you go down a group on the periodic table?
2. Explain what the following two diagrams are telling us about atomic radii.
 - (a) 
Li Be B C N O F
 - (b) 
Na Na⁺ Cl Cl⁻
3. Explain what will happen to the electronegativity of elements as you move:
 - to the right across a period of the table and
 - up a group on the table.
4. A new element is discovered and found to have 2 valence electrons. Write a paragraph describing the physical and chemical properties of this new element. What group of the Periodic Table would you place it in?
5. Bonding changes as you move across the Periodic Table from metallic to covalent network to covalent molecular. Relate the changes in bonding to the number of valence electrons present.